Roll No.: | | | | | | | | | | | |

Test Date: 02-11-2025

Aakash
Medical | IIT-JEE | Foundations

F
Booklet Code

AR-008

A I A T S
ALL INDIA AAKASH TEST SERIES

Phoenix Batch

Medical Entrance Exam - 2026

National Eligibility-cum-Entrance Test (NEET)

TEST No. 1

(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

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Test No. 1

TOPICS OF THE TEST

Physics	Units & Measurements, Motion in a Straight Line, Motion in a Plane
Chemistry	Some Basic Concepts of Chemistry, Structure of Atom
Botany	Cell: The Unit of Life, Cell Cycle and Cell Division, The Living World
Zoology	Structural Organisation in Animals – Animal Tissues, Biomolecules

AAKASH Pheonix Medical/IIT-JEE Foundations (AR-002)

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MM : 720

TEST - I

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer :

1. A body is projected obliquely from ground with speed u such that its range is maximum. Match the following physical quantities in column-I with their correct expressions given in column-II.

	Column-I		Column-II
A.	Speed at half of the maximum height	(p)	$\frac{u}{\sqrt{2}}$
B.	Speed at the maximum height	(q)	$\frac{\sqrt{3}u}{2}$
C.	Magnitude of change in velocity when it returns to the ground	(r)	$\frac{\sqrt{5}u}{2}$
D.	Average velocity till it reaches the ground	(s)	$\sqrt{2}u$

(1) A(q), B(p), C(s), D(r)

(2) A(p), B(q), C(s), D(r)

(3) A(q), B(p), C(s), D(p)

(4) A(p), B(q), C(r), D(s)

2. A body starts at $t = 0$ with velocity u and its acceleration is given by at, where a is constant. Its velocity as a function of time t is given as

(1) $u + at$

(2) $u + 2at$

(3) $ut + \frac{1}{2}at^2$

(4) $u + \frac{at^2}{2}$

3. A car starts from rest and accelerates at 3 m/s^2. At $t = 2$ s, a ball is dropped out of a window by a boy sitting in the car. The speed of the ball at $t = 4$ s will be ($g = 10$ m/s^2)

(1) $\sqrt{464}$ m/s

(2) $\sqrt{436}$ m/s

(3) $\sqrt{525}$ m/s

(4) $\sqrt{547}$ m/s

4. Plane angle has

(1) Unit but no dimension

(2) Dimension but no unit

(3) Both unit and dimension

(4) Neither unit nor dimension

5. The pitch of an error free screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a thick wire, the pitch scale reads 1 mm and 57th division on the circular scale coincides with the reference line. The diameter of the wire is

(1) 0.157 mm

(2) 1.057 mm

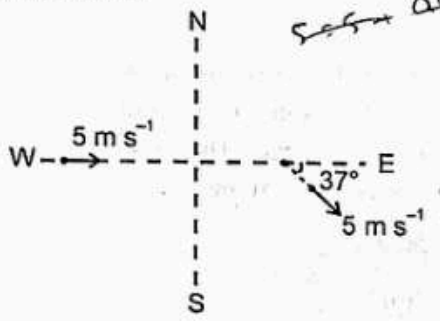
(3) 1.507 mm

(4) 1.57 mm

Space for Rough Work

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6. A particle is moving eastwards with a speed of 5 m s^{-1}. After 2 s, the particle is found to be moving with same speed in a direction 37° south of east. The magnitude of average acceleration in this interval of time is



(1) $\sqrt{\frac{2}{5}}$ m s^{-2} (2) $\sqrt{\frac{5}{2}}$ m s^{-2}

(3) $\frac{5}{2}$ m s^{-2} (4) 4 m s^{-2}

7. Three physical quantities are related as $Q = \frac{A^2}{B}$, and the percentage errors in A and Q are 3% and 2% respectively, the maximum percentage error in B is

(1) 8% (2) 6%

(3) 9% (4) 11%

8. A displacement-time graph is a straight line inclined at 30° to the time axis for a body moving on a straight line. The velocity of the body is (in SI unit)

(1) Zero (2) $\sqrt{3}$

(3) $\frac{1}{\sqrt{3}}$ (4) 1

9. A particle starts with velocity u, moves with constant retardation a, and comes to rest after sometime. The total displacement and distance covered by the particle are

(1) Both zero

(2) Both non-zero

(3) Displacement zero, distance non-zero

(4) Distance zero, displacement non-zero

10. Consider the following statements.

Statement (A) : In ground to ground projectile, if the initial velocity of projection of a body is increased n times, then its time of flight also becomes n times (θ remains same).

Statement (B) : The horizontal range of a projectile is independent of its initial velocity.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

11. A particle is moving in a circular path of radius r with decreasing speed. At a certain instant, its speed is v and it is decreasing at a rate K. The magnitude of the total acceleration of the particle is

(1) $\sqrt{K^2 + \frac{v^4}{r^2}}$ (2) $K + \frac{v^2}{r}$

(3) $\sqrt{K^2 + \frac{v^2}{r}}$ (4) $\frac{v^2}{r}$

12. A body of mass m is moving in a circular path with uniform angular velocity, the quantity that remains constant during this motion is

(1) Acceleration (2) Velocity

(3) Momentum (4) Speed

13. Two projectiles are thrown with the same speed but at complementary angles 30° and 60° from ground. The ratio of their maximum height is

(1) 1 (2) $\frac{1}{2}$

(3) $\frac{1}{3}$ (4) $\frac{1}{\sqrt{2}}$

Space for Rough Work

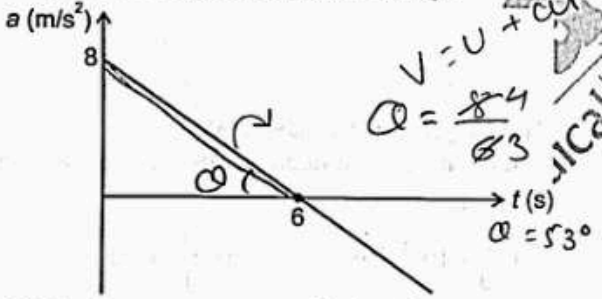
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14. A motorboat needs to reach a point located upstream at 30° w.r.t. normal (to bank), on the opposite bank of a river flowing at 5 m/s. If the boat's speed relative to water is $5\sqrt{3}$ m/s, then the angle at which the driver should steer the boat is
(1) 120° with respect to the direction of flow
(2) 150° with respect to the direction of flow
(3) 90° with respect to the direction of flow
(4) Not possible

15. A bomb is dropped from an aircraft moving horizontally at a speed of 49 m/s from a height of 490 m. The total displacement of the bomb during its flight will be (g = 9.8 m/s²)
(1) 980 m (2) 490 m
(3) 735 m (4) $490\sqrt{2}$ m

16. Two objects are moving at constant speeds. When they are moving towards each other, the distance between them closes by 6 meters every second. When they are moving in the same direction, the distance between them closes by 6 meters every 10 seconds. The speeds of the objects are
(1) 3.3 m/s, 2.7 m/s (2) 4.3 m/s, 1.7 m/s
(3) 4.3 m/s, 3.7 m/s (4) 3.3 m/s, 2.5 m/s

17. A particle is moving on a straight line according given acceleration (a)-time (t) graph. At what time the particle acquires its initial velocity?



(1) t = 9 s (2) t = 16 s
(3) t = 12 s (4) t = 15 s

18. A car is moving with an initial velocity of 108 km/h. It is brought to rest in 10 seconds by applying brakes. The distance it covers before stopping is
(1) 250 m (2) 450 m
(3) 150 m (4) 90 m

19. Which of the following statements are true about the acceleration in one dimensional motion?
(a) Positive acceleration always means the particle is speeding up.
(b) If a particle is slowing down, its acceleration must be negative.
(c) A particle at the highest point of upward motion has zero velocity but non-zero acceleration.
(d) The sign of acceleration depends on the choice of coordinate axis.
Choose the correct option.
(1) Both (a) and (c)
(2) Both (c) and (d)
(3) Both (b) and (d)
(4) Both (b) and (c)

20. A body's moment of inertia is measured as 10 kg m². If a new unit system uses 5 cm as the unit of length and 10 g as the unit of mass, then the numerical value of the moment of inertia in this new system will be
(1) 4×10^3 (2) 4×10^5
(3) 400 (4) 40

21. A gas bubble from an explosion under water oscillates with a period T proportional to $P^aD^bE^c$, where P is the pressure, D is the density of water, and E is the energy of the explosion. The value of $a + b - c$ is equal to
(1) $-\frac{1}{3}$ (2) $\frac{4}{3}$
(3) $-\frac{2}{3}$ (4) $-\frac{4}{3}$

Space for Rough Work

$V = \frac{4}{3} + (8)(6)$

$V = \frac{4}{3} + 48 = \frac{4+144}{3} = \frac{148}{3} = 49.3$

$\frac{4}{3} = U + (8)(6)$ $\frac{4}{3} - 48$ $\frac{140}{3} = 46.67$

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22. In an experiment, the refractive index of glass was measured to be 1.54, 1.56, 1.53, 1.55, 1.57. The value of mean absolute error in the measured data is
(1) 0.01 (2) 0.013
(3) 0.02 (4) 0.015

23. Consider the following statements about significant figures:
(A) The measurement 4.700 has two significant figures.
(B) In the measurement 0.00560, only three significant figures are present.
(C) Changing the unit of measurement can alter the number of significant figures.
(D) Scientific notation removes ambiguity in determining significant figures.
The correct statement(s) is/are
(1) Both (A) and (B)
(2) Both (B) and (D)
(3) Both (C) and (D)
(4) All statements are correct

24. Which of the following incorrectly describes rules for rounding off numbers with significant figures?
(1) If the digit to be dropped is more than 5, the preceding digit is raised by 1.
(2) If the digit to be dropped is 5 and the preceding digit is even, it remains unchanged.
(3) If the digit to be dropped is 5 and the preceding digit is odd, it is decreased by 1.
(4) If the digit to be dropped is less than 5, the preceding digit remains unchanged.

25. Consider the following statements:
(a) Only quantities with the same dimensions can be added or subtracted.
(b) Dimensional correctness guarantees physical correctness of an equation.
(c) Dimensionless constants cannot be determined using dimensional analysis.
The correct statement(s) is/are
(1) Only (a) (2) Both (a) and (b)
(3) Only (c) (4) Both (a) and (c)

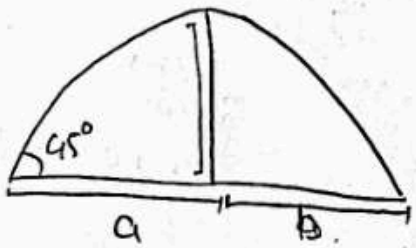
26. A projectile is launched obliquely from ground and lands back at the same level. The average velocity of the projectile during its entire flight is
(1) Zero
(2) Equal to its initial velocity
(3) Equal to its average speed
(4) Along the horizontal direction

27. From the top of a 45 meter high building, a ball is launched with an initial speed of 30 m/s at an angle of 30° above the horizontal. The total time it takes for the ball to hit the ground is ($g = 10$ m/s^2)
(1) $\frac{7-\sqrt{45}}{2}$ s (2) $\frac{3+\sqrt{45}}{2}$ s
(3) $\frac{9-\sqrt{45}}{2}$ s (4) $\frac{9+\sqrt{45}}{2}$ s

28. A ball is thrown from a point on the ground, at a horizontal distance 'a' from the base of a pole. The ball, launched at a 45° angle, grazes the top of the pole and lands on the ground at a horizontal distance 'b' (from the pole) on the pole's opposite side. The height of the pole is
(1) $\frac{ab}{a+b}$ (2) $\frac{ab}{|a-b|}$
(3) $\frac{2ab}{a+b}$ (4) $\frac{2ab}{|a-b|}$

29. The length of a seconds hand of a clock is 5 cm. The centripetal acceleration of the tip of the seconds hand, is (take $\pi^2 = 10$)
(1) $\frac{5}{9}\times10^{-3}$ m/s^2 (2) $\frac{9}{15}\times10^{-3}$ m/s^2
(3) $\frac{5}{17}\times10^{-3}$ m/s^2 (4) $\frac{9}{17}\times10^{-3}$ m/s^2

Space for Rough Work



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30. If a person can throw a projectile to a maximum horizontal distance R_{max}, the maximum height from ground to which it will rise in this condition is

(1) $\frac{R_{max}}{2}$ (2) R_{max}

(3) $\frac{R_{max}}{4}$ (4) $2R_{max}$

31. A particle moves in xy-plane such that its velocity along the x-axis remains constant at 4 m/s, while its velocity along the y-axis changes with time according to the relation $v_y = \frac{5}{2}t$ m/s, where t is time in seconds. The equation of trajectory of the particle is (Initially, particle was at origin)

(1) $64y = x^2$ (2) $16y = 5x^2$

(3) $16y = 3x^2$ (4) $64y = 5x^2$

32. A body is released from the top of a tower of height h, it reaches ground after time t. The height of the body from the ground after $\frac{t}{3}$ seconds is

(1) $\frac{8h}{9}$ (2) $\frac{h}{9}$

(3) $\frac{5h}{9}$ (4) $\frac{4h}{9}$

33. A ball is released from rest from the top of a tall building at $t = 0$. After a time interval $t = t_0$, a second ball is projected downward from the same point with an initial speed v_0. The time t (measured from the instant the first ball was dropped) at which the second ball meets the first ball, will be

(1) $\left(\frac{2v_0 - gt_0}{v_0 + gt_0}\right)\frac{t_0}{2}$ (2) $\left(\frac{2v_0 + gt_0}{v_0 - gt_0}\right)\frac{t_0}{2}$

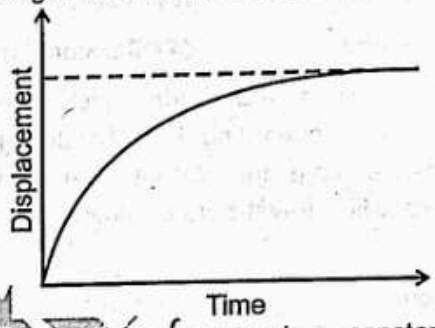
(3) $\left(\frac{2v_0 + gt_0}{2v_0 - gt_0}\right)\frac{t_0}{2}$ (4) $\left(\frac{2v_0 - gt_0}{v_0 - gt_0}\right)\frac{t_0}{2}$

34. A particle is under one dimensional motion. If speed of the particle is decreasing at an instant, then the angle between instantaneous velocity and acceleration of the particle is

(1) Zero (2) $\frac{\pi}{2}$

(3) $\frac{3\pi}{2}$ (4) π

35. The graph shows how the displacement of a particle changes with time. Based on the graph, which of the following statements is correct?



(1) The particle moves at a constant velocity throughout the motion.

(2) The particle begins with some initial velocity, but gradually slows down and eventually comes to a stop.

(3) The particle experiences constant acceleration throughout the motion.

(4) The particle starts from rest and then accelerates for some time, and finally continues with a constant velocity.

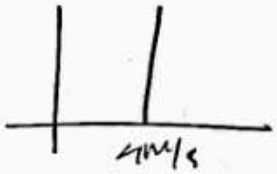
36. A particle covers half of its total distance with speed u and the rest half distance with speed v. Its average speed during the complete journey is

(1) $\frac{u+v}{2}$ (2) $\frac{2uv}{u+v}$

(3) $\frac{uv}{u+v}$ (4) $\frac{uv}{v-u}$

Space for Rough Work

$v = 4 + \frac{5}{2}t$



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37. A car accelerates uniformly from 0 to 20 m/s in 5 s, then moves at constant speed for next 10 s, and finally decelerates to come to rest in next 5 s. The total distance covered by the car in the journey is

(1) 600 m (2) 300 m

(3) 200 m (4) 175 m

38. A particle moves in a straight line with velocity $v = (6t - t^2)$ m/s, where t is time in second. It will return to its starting point after

(1) 3 second (2) 6 second

(3) 5 second (4) 9 second

39. A bullet fired in a fixed wooden block loses half of its velocity after penetrating 6 cm. Assuming the bullet experiences constant retardation, how much more distance will it travel before coming to a complete stop?

(1) 2 cm (2) 6 cm

(3) 9 cm (4) 4 cm

40. Given below are two statements, one is the assertion (A) and the other is reason (R).

Assertion (A): Universal gravitational constant (G) is a dimensional constant.

Reason (R) : Dimensional constants are unitless.

In the light of above statements, select the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

41. The number of significant figures in measurement 0.0320×10^2 m is

(1) Four (2) Five

(3) Three (4) Six

42. Instantaneous acceleration of a particle at any time t is given by $a = \alpha t^2 + \beta t$. If α and β are dimensional constants, then the factor which is dimensionless, is

(1) $\frac{\alpha t}{\beta}$ (2) $\frac{\alpha}{\beta t}$

(3) $\frac{\alpha t^2}{\beta}$ (4) $\alpha\beta t$

43. The smallest division on main scale of a vernier calliper is 1 mm and 20 vernier scale divisions coincide with 18 main scale divisions. While measuring the length of a rod, the zero mark of the vernier scale lies between 9.4 cm and 9.5 cm and 4th division of vernier scale coincide with a main scale division. Find length of the rod.

(1) 9.04 mm (2) 94.4 mm

(3) 9.4 mm (4) 94.2 mm

44. Planck's constant (h) has the same dimensions as

(1) $\frac{\text{Energy}}{\text{Frequency}}$

(2) Angular momentum

(3) Power

(4) Both (1) and (2)

45. The SI unit of Stefan's constant is

(1) W m^2 K^4

(2) W m^{-1} K^{-4}

(3) W m^{-2} K^{-4}

(4) W m K^4

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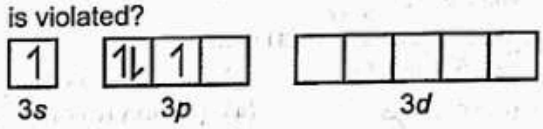
[CHEMISTRY]

46. Consider the following statements.
(a) If CH_4 and CO_2 are moving with same velocity, then de Broglie wavelength is same for both molecules.
(b) Bohr's model could not explain the ability of atoms to form molecules by chemical bonds.
(c) The angular momentum of electron in second shell of hydrogen atom is $\sqrt{2}h$.
(d) Paschen, Brackett and Pfund spectral lines for atomic hydrogen lie in Infrared region.
(e) According to Bohr's model, the most negative energy value for an electron is given by n = 1, which corresponds to the most stable orbit.
The correct statements are
(1) (a), (b), (c) and (d) only
(2) (a), (c) and (e) only
(3) (b), (d) and (e) only
(4) (a), (b), (c), (d) and (e)

47. If wavelength of the first line of Balmer series of hydrogen atom is x nm, then the wavelength of the third line of this series for hydrogen atom (in nm) will be
(1) 1.72 x (2) 0.87 x
(3) 0.66 x (4) 0.92 x

48. Consider the following statements
(a) d_{z^2} and $d_{x^2-y^2}$ have electron density along the axes.
(b) The maximum number of 3p electrons in an atom having spin quantum value of $-\frac{1}{2}$ is three.
(c) Shape of orbital is determined by Azimuthal quantum number.
(d) K^+, Ca^{2+} and S^{2-} are isoelectronic species.
(e) The number of protons, neutrons and electrons in $^{23}_{11}Na$, respectively are 11, 12 and 11.
The correct statements are
(1) (a), (b) and (c) only
(2) (a) and (e) only
(3) (c), (d) and (e) only
(4) (a), (b), (c), (d) and (e)

49. In the given configuration representation which rule is violated?



(A) Aufbau principle
(B) Pauli's exclusion principle
(C) Hund's rule of maximum multiplicity
Choose the correct option.
(1) (A), (B) and (C) (2) (A) and (C) only
(3) (B) and (C) only (4) (B) only

50. The correct decreasing order of wavelength of given electromagnetic radiations is
(1) Radio waves > Microwave > UV > IR
(2) UV > IR > Microwave > Radio waves
(3) Radio waves > Microwave > IR > UV
(4) IR > UV > Microwave > Radio waves

1. Given below are two statements.
Statement I: Values of all four quantum numbers for the valence electron of potassium can be $n = 4, l = 0, m_l = 0, m_s = +\frac{1}{2}$
Statement II: Total number of electrons in Cu which have (n + l) = 4, is 8
In the light of above statements, choose the **correct** answer from the option given below.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

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52. Consider the following statements.
(a) $^{14}_{6}C$, $^{15}_{7}N$, $^{17}_{9}F$ are isotones.
(b) In copper atom, 14 electrons have a spin of one type and 15 electrons of the opposite type.
(c) If the de Broglie wavelength of the electron in the Bohr orbit in a hydrogen like species is equal to $3\pi a_0$ (a_0 is Bohr radius), then the value of $\frac{n}{Z}$ will be 1.5.

The correct statement(s) is/are
(1) (a), (b) and (c) (2) (a) and (b) only
(3) (c) only (4) (b) and (c) only

53. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): Energies of 2*s* orbital of hydrogen atom is lower than that of 2*p* orbital of hydrogen atom.
Reason (R) : For hydrogen atom, the lower the value of (n + l) for an orbital, the lower is its energy.
In the light of above statements, choose the correct answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

54. Incorrect set of quantum numbers from the following is
(1) n = 5, l = 4, m_l = –4, –3, –2, –1, 0, +1, +2, +3, +4, $m_s = \frac{-1}{2}$
(2) n = 3, l = 2, m_l = –2, –1, 0, +1, +2, $m_s = \frac{+1}{2}$
(3) n = 2, l = 1, m_l = –1, 0, +1, $m_s = \frac{-1}{2}$
(4) n = 4, l = 4, m_l = –4, –3, –2, –1, 0, +1, +2, +3, +4, $m_s = \frac{+1}{2}$

55. The hydrogen like species He^+ ion is in a spherically symmetric state P_1 with zero radial node. Upon absorbing light the ion undergoes transition to a state P_2. The state P_2 has 1 radial node and its velocity is equal to the ground state velocity of the hydrogen atom.
Choose the correct statements
(a) The state P_1 and P_2 can be 1*s* and 2*s* respectively.
(b) Energy of the state P_2 for He^+ ion is –13.6 eV
(c) The orbital angular momentum quantum number of P_2 state is 1.
(d) The orbital angular momentum for P_2 state of He^+ ion is $\sqrt{2}\hbar$
(1) (c) and (d) only (2) (a), (b) and (c) only
(3) (a) and (b) only (4) (b) and (d) only

56. Match List-I with List-II.

	List-I		List-II Radial node(s)
a.	3*d*	(i)	0
b.	4*s*	(ii)	2
c.	4*p*	(iii)	3
d.	5*f*	(iv)	1

The correct match is
(1) a(i), b(iii), c(ii), d(iv) (2) a(ii), b(iii), c(i), d(iv)
(3) a(iii), b(ii), c(iv), d(i) (4) a(ii), b(i), c(iv), d(iii)

57. Given below are the quantum numbers for 4 electrons of a multielectronic species.
(a) n = 4, l = 2, m_l = 1, $m_s = +\frac{1}{2}$
(b) n = 4, l = 1, m_l = 0, $m_s = +\frac{1}{2}$
(c) n = 5, l = 0, m_l = 0, $m_s = -\frac{1}{2}$
(d) n = 3, l = 2, m_l = 1, $m_s = +\frac{1}{2}$

The correct order of increasing energy is
(1) (b) < (d) < (a) < (c) (2) (d) < (b) < (c) < (a)
(3) (a) < (b) < (c) < (d) (4) (a) < (c) < (b) < (d)

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58. A ball weighing 6.6 g is moving with a velocity of 80 m s^{-1}. If the uncertainty in its velocity is 5%, then the uncertainty in its position is
(1) 1×10^{-31} m (2) 4×10^{-33} m
(3) 2×10^{-33} m (4) 5×10^{-31} m

59. Given below are two statements.
Statement I: The electron in the Li^{2+} ion undergoes transition from higher orbit to orbit of radius 158.7 pm. This transition is associated with Paschen series.
Statement II: When the excited electron of a hydrogen atom drops from n = 6 to the n = 2, the maximum number of emission lines observed are 4.
In the light of above statements, choose the **correct** answer from the option given below.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

60. A 30 watt bulb emits monochromatic light of wavelength of 666 nm. The number of photons emitted per second by the bulb is
(1) 10.05×10^{23} (2) 10.05×10^{19}
(3) 10.05×10^{21} (4) 10.05×10^{22}

61. The work function of metal is 4.4×10^{-19} J. If photons of wavelength 100 nm are incident on the metal, then the maximum kinetic energy of the ejected electrons will be
(1) 15.4×10^{-19} J (2) 15.4×10^{-21} J
(3) 15.4×10^{-20} J (4) 15.4×10^{-18} J

62. Consider the following statements
(a) Frequency of emitted radiation of maximum intensity from a black body goes from a lower frequency to a higher frequency as the temperature increases
(b) Photoelectric effect and diffraction phenomenon can be explained by wave nature of electromagnetic radiation.
(c) Rutherford's α-particle scattering experiment led to the discovery of nucleus.
(d) Work function (W_0) for Li is higher than Na.
The correct statement(s) is/are
(1) (a), (c) and (d) only
(2) (a) only
(3) (b) and (c) only
(4) (b), (c) and (d) only

63. The ratio of radii of hydrogen atom in first Bohr orbit to that of radii of He^+ ion in 2nd Bohr orbit is
(1) 1 : 4 (2) 1 : 2
(3) 2 : 1 (4) 4 : 1

64. The kinetic energy of an electron in the third Bohr orbit of a Li^{2+} ion is (a_0 = Bohr radius)
(1) $\frac{h^2}{4\pi^2 m a_0^2}$ (2) $\frac{2h^2}{\pi^2 m a_0^2}$
(3) $\frac{4h^2}{\pi^2 m a_0^2}$ (4) $\frac{h^2}{8\pi^2 m a_0^2}$

65. The energy of the electron in third and fourth Bohr's orbits of the hydrogen atom is -2.41×10^{-12} erg and -1.36×10^{-12} erg respectively. The wavenumber of the emitted radiation when electron drops from the fourth Bohr's orbit to the third Bohr's orbit is
(1) 5.3×10^3 cm^{-1} (2) 2.01×10^5 cm^{-1}
(3) 2.01×10^4 cm^{-1} (4) 5.3×10^5 cm^{-1}

66. Choose the incorrect statement.
(1) $\frac{e}{m}$ ratio of electron is higher than the $\frac{e}{m}$ ratio of proton.
(2) 1s electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon.
(3) In a given electric field, β-particles are deflected less than α-particles in spite of α-particles having larger charge.
(4) The ratio of radius of atom to that of nucleus is of the order of 10^5.

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67. 20 mL of 1 M NaOH solution is added to 200 mL of 0.25 M NaOH solution. The final concentration of resultant solution will be
(1) 0.17 M (2) 0.21 M
(3) 0.28 M (4) 0.32 M

68. 40 g of an impure sample of calcium carbonate decomposes on heating to give 11.2 g calcium oxide. The percentage purity of calcium carbonate in the sample is (At wt: Ca = 40 u)
(1) 25% (2) 80%
(3) 75% (4) 50%

69. Consider the following reaction sequence:
$N_2(g) + O_2(g) \rightarrow 2NO(g)$
$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
If 14 g of $N_2(g)$ reacts with 32 g of $O_2(g)$, then maximum amount of $NO_2(g)$ produced is
(1) 23 g (2) 115 g
(3) 46 g (4) 92 g

70. When $CO_2(g)$ is passed over red hot coke it partially gets reduced to CO(g). Upon passing 1 L of $CO_2(g)$ over red hot coke, the total volume of the gases increased to 1.2 L at STP. The composition of the gaseous mixture at STP is
(1) CO_2 = 0.7 L; CO = 0.5 L
(2) CO_2 = 0.8 L; CO = 0.4 L
(3) CO_2 = 0.5 L; CO = 0.7 L
(4) CO_2 = 0.4 L; CO = 0.8 L

71. A sample of a hydrate of barium chloride weighing 122 g was heated until all the water of hydration is removed. The dried sample weighed 104 g. The molar mass of the hydrated salt is (Molar mass Ba = 137 g mol^{-1}, Cl = 35.5 g mol^{-1})
(1) 280 g mol^{-1} (2) 262 g mol^{-1}
(3) 226 g mol^{-1} (4) 244 g mol^{-1}

72. Calcium carbonate reacts with aqueous HCl to give $CaCl_2$ and CO_2 according to the reaction,
$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$
The mass of $CaCO_3$ required to react completely with 50 mL of 0.25 M HCl is
(1) 1.25 g (2) 0.312 g
(3) 0.625 g (4) 2.5 g

73. Given below are two statements.
Statement I: Molarity of 30% $\left(\frac{w}{V}\right)$ aqueous urea solution is 5 M.
Statement II: Molality of 30% $\left(\frac{w}{w}\right)$ aqueous urea solution is 7.14 m.
In the light of above statements, choose the **correct** answer from the options given below.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

74. Match List-I with List-II.

	List-I Species		List-II Number of electrons
a.	4 g of CH_4	(i)	$2N_A$
b.	2 g of SO_2	(ii)	$4N_A$
c.	4 g of CO_2	(iii)	$2.5N_A$
d.	8 g of NO_2	(iv)	N_A

The correct match is
(1) a(i), b(ii), c(iv), d(iii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iii), b(ii), c(i), d(iv) (4) a(iii), b(iv), c(i), d(ii)

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75. The ratio of the wavenumber of light absorbed by a Li^{2+} ion when it undergoes $n = 2 \rightarrow n = 3$ to $n = 5 \rightarrow n = 6$ transitions, is

(1) 11 : 25

(2) 36 : 25

(3) 125 : 11

(4) 25 : 36

76. 2 moles of PQ weigh 24×10^{-3} kg and 4 moles of PQ_2 weigh 56×10^{-3} kg. The molar mass of P(M_P) and molar mass of Q(M_Q) in kg mol^{-1} are

(1) $M_P = 1 \times 10^{-2}$ and $M_Q = 2 \times 10^{-3}$

(2) $M_P = 2 \times 10^{-2}$ and $M_Q = 4 \times 10^{-4}$

(3) $M_P = 2 \times 10^{-3}$ and $M_Q = 1 \times 10^{-2}$

(4) $M_P = 4 \times 10^{-4}$ and $M_Q = 2 \times 10^{-2}$

77. Given below are two statements.

Statement I: If an aqueous solution of 12.6 g oxalic acid dihydrate is made upto 1000 mL, then the volume of 0.1 N NaOH required to completely neutralize 10 mL of this oxalic acid dihydrate solution will be 20 mL.

Statement II: A gaseous mixture contains nitrogen gas and oxygen gas in the ratio of 1 : 2 by weight. Therefore, the ratio of moles of nitrogen gas to oxygen gas will be 4 : 7.

In the light of above statements, choose the **correct** answer from the option given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

78. If 200 mL of 0.2 M $BaCl_2$ is mixed with 500 mL of 0.1 M Na_3PO_4, the maximum amount of $Ba_3(PO_4)_2$ that can be formed is

(Atomic weight of Ba = 137 u)

(1) 17.02 g (2) 9.08 g

(3) 8.01 g (4) 18.06 g

79. The amount of zinc required to produce 448 mL of $NO_2(g)$ at STP on treatment with concentrated HNO_3 will be (Molar mass of Zn = 65 g/mol)

$Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2(g)$

(1) 0.065 g (2) 0.65 g

(3) 1.3 g (4) 0.13 g

80. During the Ostwald's process, if 34 g of NH_3 gas reacts with 44.8 L of $O_2(g)$ at STP, then consider the following statements.

$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$

(a) O_2 is limiting reagent

(b) 48.0 g of NO(g) is formed

(c) 143.36 L gaseous product present at the end of reaction at STP

(d) 53.76 L of H_2O (g) is formed at STP

The correct statements are

(1) (a) and (d) only (2) (b), (c) and (d) only

(3) (a), (b) and (d) only (4) (b) and (c) only

81. An organic compound is composed of 50% oxygen, 12.5% hydrogen and 37.5% carbon by mass, what would be the empirical formula of the compound?

(1) C_2H_2O (2) CH_2O_2

(3) CH_4O (4) $C_2H_4O_2$

82. An oxide of a metal contains 32% oxygen, by weight. The equivalent weight of the metal is

(1) 17 (2) 32

(3) 8 (4) 34

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83. SO_2Cl_2 on reaction with excess of water results into acidic mixture.

$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$

20 moles of NaOH is required for the complete neutralisation of the resultant acidic mixture. The number of moles of SO_2Cl_2 used is

(1) 4 (2) 20
(3) 8 (4) 5

84. Sodium oxide reacts with water to produce sodium hydroxide. 1.55 g of sodium oxide is dissolved in 250 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is

(1) 0.1 M (2) 0.2 M
(3) 0.3 M (4) 0.4 M

85. A protein 'A' contains 0.20% of alanine (molecular weight = 89 u). The minimum molar mass of protein 'A' is

(1) 4.45×10^2 g/mol (2) 2.12×10^3 g/mol
(3) 4.45×10^4 g/mol (4) 2.12×10^4 g/mol

86. The maximum amount of $O_2(g)$ consumed per gram of reactant is for the reaction

(1) $2Ca + O_2 \rightarrow 2CaO$
(2) $B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$
(3) $S_8(s) + 8O_2(g) \rightarrow 8SO_2(g)$
(4) $C(s) + O_2(g) \rightarrow CO_2(g)$

87. Choose the incorrect match

	Base physical quantity		Name of SI unit
(1)	Amount of substance	–	Kilogram
(2)	Length	–	Metre
(3)	Time	–	Second
(4)	Luminous intensity	–	Candela

88. Choose the pure substances among the following.

(a) Milk
(b) Mud
(c) Alloy
(d) Ammonia
(e) Gold

Choose the correct answer from the options given below.

(1) (d) and (e) only (2) (b), (c) and (d) only
(3) (a), (b) and (c) only (4) (a) and (d) only

89. Match List-I with List-II.

	List-I		List-II Significant figure
a.	0.03	(i)	3
b.	285	(ii)	Infinite
c.	0.0050	(iii)	1
d.	20 eggs	(iv)	2

The correct match is

(1) a(iii), b(i), c(iv), d(ii) (2) a(iv), b(ii), c(iii), d(i)
(3) a(ii), b(iii), c(i), d(iv) (4) a(i), b(iii), c(ii), d(iv)

90. Among the following, choose the ones with equal number of atoms.

(A) 22.8 g of N_2O_3 (B) 16 g of O_3
(C) 20 g of NaOH (D) 22 g of CO_2
(E) 49 g of H_2SO_4

Choose the correct answer from the options given below.

(1) (A), (B) and (C) only
(2) (B), (D) and (E) only
(3) (B), (C) and (E) only
(4) (A), (C) and (D) only

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[BOTANY]

91. Select the **correct** statement w.r.t. cytokinesis.
(1) In an animal cell, it is achieved by the centrifugal growth of cell furrow.
(2) In a plant cell, phragmoplast is formed by Golgi complex to form cell plate.
(3) Cytokinesis divides the nucleus into two equal halves.
(4) Furrow formation starts at the periphery and then moves inwards, dividing the cell into two halves in plants.

92. Amongst the following, the stage of cell division which is typically longer and more complex as compared to the others is
(1) Telophase of meiosis II
(2) Prophase of meiosis I
(3) Prophase of meiosis II
(4) Metaphase of meiosis I

93. Which of the following phases leads to the reduction in the number of chromosomes in daughter cells?
(1) Anaphase of mitosis (2) Anaphase I
(3) Anaphase II (4) Telophase

94. In oocytes of some vertebrates, which stage lasts for months or years?
(1) Metaphase I (2) Anaphase II
(3) Diakinesis (4) Dictyotene

95. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	Zygotene ii	(i)	Chromosomes gradually become visible under the light microscope
(b)	Pachytene iv	(ii)	Homologous chromosomes start pairing together and this process leads to the formation of bivalent
(c)	Leptotene i	(iii)	Represents the transition to metaphase
(d)	Diakinesis iii	(iv)	Recombination between homologous chromosomes is completed by the end of this stage.

	a	b	c	d
(1)	ii	i	iv	iii
(2)	ii	iv	i	iii
(3)	iii	ii	iv	i
(4)	iv	ii	i	iii

96. Read the following statements and identify them as true (T) or false (F) and accordingly choose the **correct** option.
A. The final stage of meiotic prophase I is marked by terminalisation of chiasmata.
B. Nucleolus, Golgi complex and ER reform during the end of prophase.
C. The cells of the upper layer of the epidermis and the cells of the lining of gut, get constantly replaced through meiosis.
D. Mitosis usually results in the production of diploid daughter cells with identical genetic complement.

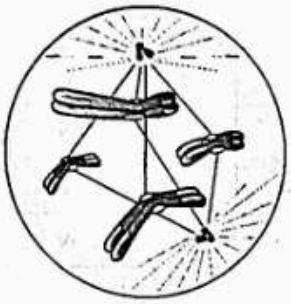
	A	B	C	D
(1)	T	T	F	T
(2)	T	F	T	F
(3)	F	T	F	T
(4)	T	F	F	T

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97. Dissolution of the synaptonemal complex in meiotic cell division occurs during
(1) Leptotene (2) Diakinesis
(3) Pachytene (4) Diplotene

98. A stage of cell division is shown in the following figure. Select the answer which gives the **correct** identification of the stage with its characteristics.



(1)	Late anaphase	–	Chromosomes move away from the equatorial plate and Golgi complex is absent
(2)	Cytokinesis	–	Cell plate is formed and mitochondria are distributed between two daughter cells
(3)	Transition to metaphase	–	Chromosomes are moving to the spindle equator and to get aligned along the metaphase plate
(4)	Prophase	–	The two asters, together with spindle fibres, form mitotic apparatus

99. During which stage of meiosis, tetrad of chromosomes begins to appear clearly?
(1) Leptotene (2) Zygotene
(3) Pachytene (4) Prophase

100. Various events of meiosis are given below.
(a) Reappearance of nuclear membrane around chromosomal monads and nucleolus.
(b) Duplication of centrioles in animal cell along with RNA and protein synthesis.
(c) Chromosome are fully condensed and the meiotic spindle is assembled.
(d) Homologous chromosomes separate and sister chromatids remain associated at their centromere.
(e) The bivalent chromosomes align on the equatorial plate.

Select the **correct** sequence of the events of meiosis from the following options.
(1) (a), (b), (d), (e), (c) (2) (c), (d), (e), (a), (b)
(3) (c), (e), (d), (a), (b) (4) (c), (e), (d), (b), (a)

101. Read the following assertion (A) and reason (R) statements and select the **correct** option.

Assertion (A): The mechanism during reductional division in cells which is catalysed by the enzyme recombinase introduces new combination of traits among the gametes of an organism.

Reason (R): Crossing over occurs between non-sister chromatids of homologous chromosomes during a sub-stage of prophase I.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

102. Select the **incorrectly** matched pair.

(1)	G_1 phase	–	Cell is metabolically active and most of the cell organelles duplicate
(2)	G_2 phase	–	Period of cytoplasmic growth and protein synthesis
(3)	S phase	–	DNA replication begins in the nucleus and thus chromosome number increases
(4)	G_0 phase	–	Cells are metabolically active but no longer proliferate

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103. The phase during which duplicated centrosome begins to move towards opposite poles is also characterised by
(1) Splitting of centromere and movement of daughter chromatids to opposite poles
(2) The initiation of condensation of chromosomal material
(3) Decondensation of chromosomes and reformation of membrane bound organelles
(4) Attachment of spindle fibre to the kinetochore of the homologous chromosomes

104. The complete disintegration of the nuclear envelope marks the start of the phase where
(a) Chromosomes are spread through the cytoplasm of the cell.
(b) Condensation of chromosomes is completed and they can be observed clearly under the microscope.
(c) The sister chromatids reach to their respective poles.
(d) Chromosomes are moved to spindle equator and get aligned along metaphase plate.
(e) Microtubules attach to the kinetochore of chromosomes.
Select the **correct** ones.
(1) (a), (b) and (c) (2) (b), (c) and (e)
(3) (a), (b), (d) and (e) (4) (c), (d) and (e)

105. Read the following statements and select the **correct** option.
Statement A: In contrast to mitosis, meiosis occurs in the diploid cells, which are destined to form gametes.
Statement B: In coconut, liquid endosperm is formed because during its formation cytokinesis is not followed by karyokinesis.
(1) Both statements are incorrect
(2) Both statements are correct
(3) Only statement B is correct
(4) Only statement A is correct

106. The first stage of karyokinesis of mitosis is
(1) Anaphase
(2) Telophase
(3) Metaphase
(4) Prophase

107. Among eukaryotes, chromosome duplication occurs during the
(1) G_1 phase
(2) S phase
(3) G_2 phase
(4) G_0 phase

108. Most active stage of the cell cycle is
(1) A long, non-dividing, growing phase of the cell cycle where the cell prepares itself for division.
(2) Also the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell.
(3) The stage which starts with the nuclear division and ends with the division of cytoplasm.
(4) The stage that lasts for less than 5% of the duration of cell cycle.

109. Which of the following sequences regarding different phase of a single cell cycle is acceptable?
(1) Cytokinesis → Karyokinesis → Gap 1 phase → Quiescent stage. → Gap 2 phase.
(2) Gap 1 phase → Gap 2 phase → Synthesis phase → Cytokinesis.
(3) Karyokinesis → Cytokinesis → Gap 2 phase → Quiescent stage
(4) Quiescent stage → Gap 1 phase → Synthesis phase → Gap 2 phase → M phase

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110. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	Kinetochores	(i)	A part of chromosome beyond the non-staining secondary constriction.
(b)	Chromatin	(ii)	Appears as a narrow region, called primary constriction of the chromosome.
(c)	Centromere	(iii)	Composed of DNA, some basic proteins, some non-histone proteins and RNA.
(d)	Satellite	(iv)	Disc-shaped structures at primary constriction that get attached with the microtubules during cell division.

	a	b	c	d
(1)	iv	iii	ii	i
(2)	iv	ii	iii	i
(3)	iii	i	iv	ii
(4)	ii	iv	i	iii

111. Which of the following statements regarding Golgi apparatus is **incorrect**?

(1) It consists of many flat, disc-shaped sacs or cisternae of 0.5 µm to 1.0 µm diameter.

(2) The cisternae towards the convex surface form the *cis* face.

(3) The *cis* and *trans* faces of the organelle are entirely different, but interconnected.

(4) A number of proteins synthesised by ribosomes on ER are modified in the cisternae and released from forming face.

112. Cylindrical structures in the centrosome

(a) Possess a whorl of nine evenly spaced peripheral fibrils of tubulin.

(b) Possess a rod-shaped proteinaceous mass in the centre.

(c) Have an organisation like the cartwheel.

(d) Help in the formation of basal bodies which give rise to cilia and flagella.

(e) Are surrounded by plasma membrane and pericentriolar satellites.

The **correct** ones are

(1) Only (a), (b) and (e)

(2) Only (c), (d) and (e)

(3) Only (b), (c) and (d)

(4) Only (a), (b), (c) and (d)

113. Select the **wrong** statement w.r.t. microfilaments.

(1) They are solid, unbranched, rod-like fibrils of indefinite length.

(2) They provide support to plasma membrane.

(3) They help in anaphasic movement of chromosomes.

(4) They represents the contractile system of the cell and are involved in cytoplasmic streaming.

114. Which organelle is rich in almost all types of hydrolytic enzymes?

(1) Ribosome (2) Glyoxysome

(3) Lysosome (4) Nucleolus

115. The endoplasmic reticulum is composed of all the following structures, **except**

(1) Long flattened, parallel, sac-like interconnected structures

(2) Branched or unbranched structures, forming the reticular system that are free of ribosomes

(3) Oval, membrane-bound, vacuolar structures that are free of ribosomes

(4) Extremely minute, fibrous, filamentous and tubular, proteinaceous structures forming the structural framework of the cell

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116. State true (T) or false (F) for the given statements and select the **correct** option.
(a) Gas vacuoles are found in blue-green and purple and green photosynthetic bacteria.
(b) The fluid nature of the membrane is important from the point of view of functions like cell growth and cell division.
(c) The two membranes of mitochondrion have their own specific enzymes associated with the mitochondrial function.
(d) Majority of the chloroplasts of the green plants are found in the epidermal cells of the leaves.

	(a)	(b)	(c)	(d)
(1)	T	F	T	T
(2)	F	T	T	F
(3)	T	T	F	T
(4)	T	T	T	F

117. Find the **incorrect** match from the following.
(1) Thylakoids – Flattened membranous sacs in the stroma of plastids
(2) Crista – Infolding of inner membrane in mitochondrion
(3) Cisternae – Associated with ribosomes in endoplasmic reticulum
(4) Centromere – Prevents DNA loss at both the terminals of a chromosome

118. In human beings, the membrane of the erythrocyte has approximately ___A___ percent proteins and ___B___ percent lipids.
Select the **correct** option to fill in the blanks A and B respectively.
(1) 40 and 50 (2) 42 and 52
(3) 52 and 40 (4) 68 and 30

119. Select the **odd** one out w.r.t. structures included in endomembrane system of the cell.
(1) Golgi apparatus
(2) Endoplasmic reticulum
(3) Vacuole
(4) Chloroplast

120. Reserve material in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies such as
(1) Phosphate granules (2) Polyribosome
(3) Lysosomes (4) Centrosomes

121. Read the following statements of Assertion (A) and Reason (R) and select the correct option.
Assertion (A): Larger and more nuclei are required in the cells for actively carrying out protein synthesis.
Reason (R): Spherical structure present in the nucleoplasm is the site for active rRNA synthesis.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

122. Read the following statements and select the **correct** option.
Statement A: Endoplasmic reticulum helps in the transport of substances and synthesis of proteins.
Statement B: Cell wall helps in cell-to-cell interaction and provides barrier to undesirable macromolecules.
(1) Both statement A and statement B are incorrect
(2) Both statement A and statement B are correct
(3) Only statement A is correct
(4) Only statement B is correct

123. The layer of bacterial cell envelope that provides gummy and sticky character is chiefly made up of
(1) Polysaccharide macromolecules
(2) Lipoproteins
(3) Phospholipids
(4) Glycolipids and proteins

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124. Plant cells and animal cells are different as the former possess
(1) Mitochondria, cytoplasm and golgi apparatus
(2) Cell wall, plastids and a large central vacuole
(3) Centrioles, microtubules and microvilli
(4) Peroxisome, lysosome and nucleus

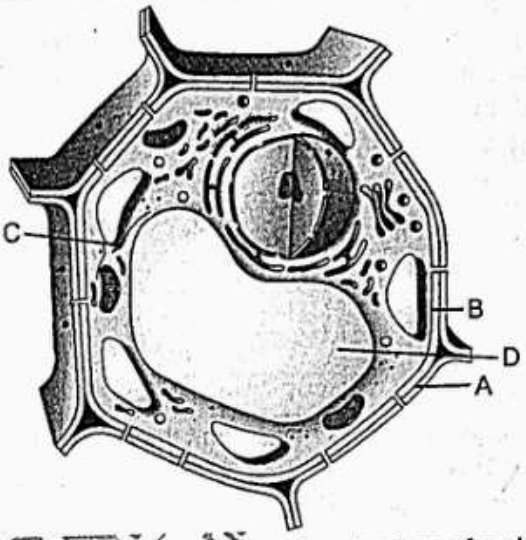
125. Which of the given structures act together as a single protective unit in bacteria?
(1) Filament, hook and basal body
(2) Pili, fimbriae and cilia
(3) Glycocalyx, cell wall and plasma membrane
(4) Mesosome, polysome and ribosome

126. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	β-oxidation of fats	(i)	It is done by non-membranous, dense particles composed of RNA and proteins, present in the cytoplasm as well as in some organelles.
(b)	Protein synthesis	(ii)	It is done by a network or reticulum of tiny tubular structures, scattered in the cytoplasm.
(c)	Lipoprotein synthesis	(iii)	It is done by a double membrane bound organelle that contains small, double stranded circular DNA and ribosomes.
(d)	Carbohydrate synthesis	(iv)	It is done by sausage-shaped, double membranous organelles, distributed in the cytoplasm.

	a	b	c	d
(1)	ii	i	iii	iv
(2)	iv	iii	i	ii
(3)	iii	iv	i	ii
(4)	iv	i	ii	iii

127.



The above figure shows the structure of a plant cell with its four parts labelled as (A), (B), (C) and (D). Select the part that has been **correctly** matched with its function.

(1)	(A): Secondary cell wall	–	Glues the different neighbouring cells together and mainly composed of lignin.
(2)	(B): Plasma membrane	–	Allows many molecules to move briefly across it without any requirement of energy.
(3)	(C): Microfilament	–	Forms cell plate during cell division by synthesising cellulosic material.
(4)	(D): Contractile vacuole	–	Contains water, sap, excretory products and other materials, useful for the cell.

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128. Who first explained that cells divide and new cells are formed from the pre-existing cells?
(1) Antonie Von Leeuwenhoek
(2) Rudolf Virchow
(3) Theodore Schwann
(4) Robert Hooke

129. In 1839, Matthias Schleiden, a German botanist, examined a large number of plants and
(1) Observed that all plants are composed of different kinds of cells which form the tissues of the plants
(2) Concluded that presence of cell wall is a unique character of the plant cells
(3) Observed that the cells had a thin outer layer, which is today known as the plasma membrane
(4) Concluded that the non-membrane bound cell organelles were unique to eukaryotes only

130. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	Plasmid	(i)	It is a region where the genetic material of bacteria is present
(b)	Capsule	(ii)	It is made up of poly-saccharides but may sometimes contain proteins also
(c)	Nucleoid	(iii)	It determines the shape of the cell and provides a strong structural support to prevent the bacterium from bursting
(d)	Cell wall	(iv)	It is a small circular DNA that confers certain unique phenotypic characters to bacteria

	a	b	c	d
(1)	iv	iii	i	ii
(2)	iv	ii	i	iii
(3)	iii	iv	ii	i
(4)	i	ii	iv	iii

131. Select the **odd** one w.r.t. cell organelles.
(1) Endoplasmic reticulum
(2) Golgi complex
(3) Peroxisome
(4) Centrosome

132. Nomenclature is governed by certain universal rules. Which one of the following is contrary to the rules of binomial nomenclature?
(1) The first word in a biological name represents the genus, while the second component denotes the specific epithet
(2) When written by hand, both the words in a biological name are to be underlined separately
(3) The names are generally in Latin and are italicised
(4) The first word of the name starts with a small letter, while the second word starts with a capital letter

133. Correct binomial epithet of mango, given by Carolus Linnaeus is
(1) *Mangifera indica* Linn.
(2) *Mangifera indica Linn.*
(3) *Mangifera indica*
(4) Mangifera indica L.

134. *Panthera* and *Felis* fall under which of the given taxonomic categories?
(1) Canidae
(2) Mammalia
(3) Primata
(4) Non-chordates

135. All the living organisms
(1) Have chemical composition identical to each other
(2) Have all the characteristics of life
(3) Share the common genetic material, but to varying degrees
(4) Are made up of cells having identical functions

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[ZOOLOGY]

136. In all of the given organisms, functions like digestion, respiration and reproduction are performed by a single cell, **except**
(1) *Amoeba* (2) *Paramoecium*
(3) *Hydra* (4) *Euglena*

137. The type of epithelial tissue which is involved in functions like forming a diffusion boundary is characterised by the presence of
(1) More than one layer of cells
(2) Loosely packed cells with abundant intercellular matrix
(3) Columnar cells which is specialised for providing protection
(4) A single layer of flattened cells

138. The type of connective tissue that has fibroblasts and many fibres (mostly collagen) oriented differently, is present in the
(1) Tendons which attach muscles to bones
(2) Ligaments which attach one bone to another
(3) Skin
(4) Main circulating fluid

139. The main function of the epithelium that forms the inner lining of ducts of salivary glands is to
(1) Provide protection against chemical and mechanical stresses
(2) Help in secretion and absorption
(3) Assist in transportation of particles in a specific direction
(4) Move mucus in any direction

140. Choose the odd one w.r.t the structures that possess heterocyclic rings.
(1) Ribose (2) Cholesterol
(3) Guanine (4) Cytosine

141. Complete the analogy by selecting the correct option.
Drug : Curcumin : : Alkaloid : _____
(1) Abrin (2) Morphine
(3) Cellulose (4) Ricin

142. The type of biomacromolecules whose monomeric unit is amino acid constitutes what % of the total cellular mass?
(1) 10-15 (2) 5-7
(3) 70-90 (4) 20-30

143. 'X' is a component that forms an integral part of a nucleotide but is absent in lecithin. Choose the option which correctly identifies 'X'.
(1) Fatty acids (2) Hexose sugar
(3) A phosphate group (4) A monosaccharide

144. The type of epithelium that is present in the inner surface of fallopian tube is also present in the structure that
(1) Aids in digestion of food
(2) Is found in the lining of human respiratory system
(3) Participates in the exchange of gases between blood and atmospheric air
(4) Helps to maintain the pH and ionic balance of the body fluid by secretion of H^+ in PCT

145. Consider the given chain
Proline → Valine → Lysine → Serine → Glycine
Select the **correct** option w.r.t. it.
(1) The right end of the above oligosaccharide chain is called the reducing end
(2) The left end of the above oligosaccharide chain is called the non-reducing end
(3) Proline is called the N-terminal amino acid
(4) Glycine is called the N-terminal amino acid

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Pheonix Medical (AESL) Batches

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146. Consider the following:
(a) A type of specialised connective tissue
(b) Present in the outer ear joints
(c) Possess solid and pliable matrix
All of the above given features are true for the type of connective tissue which is characterised by the presence of
(1) Osteocytes (2) Lamellae
(3) Chondrocytes (4) Bone marrow

147. Select the **correct** match.

(1)	Intercalated discs	–	A type of tight junction present in cardiac muscles
(2)	Nissl's granules	–	Present in cyton and dendrites
(3)	Neuroglial cells	–	Structural and functional unit of neural tissue
(4)	Adipocytes	–	Abundant in dense regular connective tissue

148. Choose the **correct** option w.r.t the type of loose connective tissue present mainly beneath the skin but is not specialised to store fats.
(1) Does not contain collagen fibres
(2) Serves as support framework for epithelium
(3) Has orientation of fibres in a regular pattern
(4) Is the main tissue that provides structural frame to the body

149. The type of muscle fibres which are present in the wall of stomach and intestine are characterised by the presence of
(a) Cylindrical and elongated shape
(b) Communication junctions
(c) Centrally placed nucleus
(d) Syncytial condition
Select the option with the correct set.
(1) a, b, c and d (2) a and b only
(3) a and d only (4) b and c only

150. Given below are two statements, one is the **Assertion (A)** and the other is **Reason (R).**
Assertion (A): Neural tissues are the most abundant and widely distributed tissue in the body of complex animals.
Reason (R) : Neural tissue connects various tissues and organs and innervate all cells of the body.
In the light of above statements, select the most appropriate option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

151. Read the following statements.
Statement A: Oxaloacetate closely resembles the structure of succinic dehydrogenase and illustrates the example of competitive inhibition.
Statement B: In case of competitive inhibition, the apparent K_m value increases while the value of V_{max} remains the same.
Select the **correct** option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is incorrect
(4) Only statement B is incorrect

152. If a comparison is made between diglyceride and triglyceride, then
(1) Both of them have only two fatty acids
(2) Latter possesses three glycerol molecules while former possesses two glycerol molecules
(3) Both of them have one glycerol moiety
(4) Former one possesses two fatty acids while latter one possesses three glycerol molecules.

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153. Suppose the E.C. number of an enzyme is 4.2.1.1. Identify the class to which this enzyme belongs to and choose the **correct** option w.r.t. it.
(1) Catalyses removal of groups from substrates by mechanisms other than hydrolysis
(2) Catalyses hydrolysis of ester, ether and peptide bonds
(3) Responsible for the linking together of two compounds
(4) Catalyses oxidoreduction between two substrates

154. Given below are three statements:
(a) Tyrosine is an aromatic amino acid which is basic in nature
(b) Water is the most abundant chemical in living organisms
(c) A common feature between insulin and trypsin is that both of them are heteropolymers
Choose the option that includes the correct statement(s) only.
(1) (a), (b) and (c) (2) (a) and (b) only
(3) (b) and (c) only (4) (b) only

155. GLUT-4 is a transporter biomacromolecule that enables glucose transport into cells. GLUT-4 belongs to the category of biomolecule 'P' that possesses monomers which are adjacently connected by bonds formed as a result of _____ reaction.
Select the option that **correctly** fills the blank.
(1) Oxidoreduction (2) Redox
(3) Dehydration (4) Hydration

156. 'X' is strictly not a macromolecule, yet it is present in acid-insoluble fraction obtained upon chemical analysis of a living tissue. 'X' constitutes what % of the total cellular mass?
(1) 10-15 (2) 5-7
(3) 2 (4) 3

157. The teacher explained to the class about different types of cell junctions which are present in the human body. Choose the **incorrect** option regarding cell junctions.
(1) The tight junctions play a crucial role in preventing leakage
(2) Three types of cell junctions are found in the epithelium and other tissues as well.
(3) Adhering junctions cement neighbouring cells together.
(4) Gap junctions help to stop substances from leaking across a tissue.

158. The cells of areolar tissue that also participate in mediating allergy is
(1) Macrophages (2) Mast cells
(3) Megakaryocytes (4) Monocytes

159. If a comparison is done between glucagon and glycogen, then which of the following statements is true?
(1) Both of them act as storage polysaccharides
(2) Former is a homopolymer while latter is a heteropolymer
(3) Former contains peptide bonds while latter contains glycosidic bonds
(4) Former possesses only left hand helices while latter possesses right handed helices

160. Consider the given statements.
Assertion (A): Enzymes differ from inorganic catalysts w.r.t. several factors such as temperature.
Reason (R) : Inorganic catalysts work efficiently at high temperatures while enzymes usually get damaged at high temperatures.
In the light of above statements, select the most appropriate option.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

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161. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Palmitic acid	(i)	Unsaturated fatty acid with C = C bonds
b.	Aspartic acid	(ii)	Saturated fatty acid without C = C bonds
c.	Acetic acid	(iii)	Possess extra carboxyl group
d.	Arachidonic acid	(iv)	Involved in cholesterol synthesis

(1) a(ii), b(iii), c(iv), d(i) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iii), b(i), c(iv), d(ii) (4) a(iv), b(iii), c(ii), d(i)

162. Choose the **correct** option w.r.t. prosthetic group of catalase.
(1) Its association with the apoenzyme is only transient
(2) It forms only coordination bonds with side chain at the active site of enzyme
(3) It is inorganic in nature
(4) It is tightly bound to the apoenzyme

163. The K_m value of an enzyme catalysed reaction denotes the substrate concentration at which the enzyme catalysed reaction
(1) Does not proceed further
(2) Reaches V_{max}
(3) Attains half of the V_{max}
(4) Exceeds beyond $V_{max}/2$

164. Tendons and ligaments can be differentiated from each other as
(1) Only former has regular orientation of fibres
(2) Only latter has compact packaging of fibres and fibroblasts
(3) Only former attach muscles to bones
(4) Only latter exhibits the property of excitability

165. The fibres which are present in most of the connective tissue
(1) Are responsible for providing strength to the tissue
(2) Are made up of modified lipids and polysaccharides
(3) Exhibit excitability and facilitate conduction of impulses
(4) Cannot regenerate due to avascular nature of connective tissue

166. The common feature(s) between cardiac muscles and skeletal muscles is/are
(a) Presence of actin and myosin filaments
(b) Autorhythmicity
(c) Presence of gap junctions
(d) Cylindrical muscles fibres
Choose the **correct** option.
(1) (a) and (b) (2) (c) and (d)
(3) Only (d) (4) (a) and (d)

167. Choose the **odd** one w.r.t. the location of a specialised connective tissue in adults whose matrix is solid and pliable and resists compression.
(1) Between adjacent thoracic vertebrae
(2) A flap that covers glottis
(3) A structure that helps in sound production
(4) Between cranial bones

168. Match column I with column II and choose the correct option.

	Column I		Column II
a.	Phenylalanine	(i)	4 homocyclic rings
b.	Adenine	(ii)	3 heterocyclic rings
c.	Guanylic acid	(iii)	2 heterocyclic rings
d.	Cholesterol	(iv)	1 homocyclic ring

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(ii), d(i) (4) a(iv), b(i), c(iii), d(ii)

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Pheonix Aakash Medical/IIT-JEE (AIATS)

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169. All of the following are present in the main circulating fluid of the body that helps in the transport of various substances, **except**

(1) Erythrocytes

(2) Thrombocytes

(3) Macrophages

(4) Leucocytes

170. 'P' make up more than one-half the volume of the neural tissue in our body. Identify 'P' and select the **correct** option w.r.t. it.

(1) Assist in conduction of nerve impulses

(2) Provide protection and support to the neurons

(3) Receive and transmit signals

(4) Involved in production of stimuli

171. Read the following statements and choose the **correct** option.

Statement A: Lysine and valine are considered as basic amino acids as they possess an extra amino group in their structures.

Statement B: In solutions of different pH, the structure of amino acids changes due to the ionisable nature of $-NH_2$ and $-COOH$ groups.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

172. Choose the correct statement w.r.t. an adult human.

(1) Arrival of the disturbance at the neuron's endings trigger events that cause only stimulation of adjacent neurons.

(2) All the bones contain the site of production of blood cells.

(3) Muscle fibres lengthen in response to stimulation and then shorten to their uncontracted state.

(4) Neural tissue exerts the greatest control over the body's responsiveness to changing conditions.

173. The total number of nitrogen atoms present in the ring of uracil is equal to the number of H-bonds present between

(1) Guanine and cytosine

(2) Adenine and thymine

(3) Trihydroxypropane and choline

(4) Monomeric units of arachidonic acid

174. The class of enzymes that catalyse the formation of a phosphodiester bond between two nucleotides for joining of two separate DNA strands is

(1) III (2) IV

(3) VI (4) I

175. Select the **incorrect** match.

(1)	Chitin	–	Exoskeleton of arthropods
(2)	Starch	–	Forms helical secondary structures
(3)	Cellulose	–	Polymeric secondary metabolite
(4)	Cytosine	–	Present exclusively in RNA

176. How many of the biomolecules mentioned in the box below are characterised by the presence of peptide bonds?

Collagen, RuBisCO, Glycogen, Inulin, Trypsin, Antibody

Choose the **correct** option.

(1) Four (2) Three

(3) Five (4) Two

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177. Consider the following
(a) Alpha helix
(b) Presence of hydrogen bonds
(c) Presence of phosphodiester bonds
(d) β-pleated sheet
Which of the above is correct w.r.t. secondary structure of proteins?
(1) (a), (b), (c) and (d) (2) (b), (c) and (d) only
(3) (a), (b) and (d) only (4) (a), (c) and (d) only

178. Select the **incorrect** statement.
(1) Goblet cells of the alimentary canal are modified columnar glandular cells.
(2) The epithelial tissue has a free surface which always faces the outside environment.
(3) Salivary gland is an example of multicellular gland that consists of cluster of cells.
(4) In nearly all animal tissues, specialised junctions provide both structural and functional links between its individual cells.

179. _____ is the most abundant protein in the animal world while _____ is the most abundant protein in the whole of the biosphere.
Fill in the respective blanks by selecting the **correct** option.
(1) Cellulose ; Collagen
(2) Collagen ; RuBisCO
(3) RuBisCO ; Collagen
(4) Collagen ; Cellulose

180. A student analysed the different categories of connective tissues and he compared blood, areolar tissue and adipose tissue. After the comparison, he found that
(1) All of them possess fibroblasts
(2) Blood possesses macrophages while adipose tissue lacks them
(3) Both blood and areolar tissue possess mast cells
(4) Both areolar tissue and adipose tissue possess adipocytes

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